NAME: ____ CLASS: _________ INDEX: __________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

ANSWERS

BIOLOGY
9744 / 04

Paper 4 Practical

6th / 7th Aug 2018

2 hours 30 minutes

Candidates answer on the Question Paper

Additional Materials:

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
TOTAL	

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **12** printed pages and **0** blank page.

[Turn over

[Turn over

Answer **all** questions

QUESTION 1

Diarrhoea in babies can be a serious problem that can have several different causes. One such cause is carbohydrate intolerance, usually stemming from a genetic condition which results in a very low level of a biomolecule **K**, which helps break down starch and sucrose in the intestine.

P is an extract of intestinal juices from a baby patient which needs to be tested for its concentration of **K**. You are tasked to help determine the concentration of the biomolecule **K** in extract **P**.

You are provided with the following:

1. The sample of intestinal juice from patient labelled **P**.
2. 10 cm^3 solution labelled **K1** containing 5.0% **K**.
3. 20 cm^3 solution labelled **S** containing 0.5% starch.
4. Distilled water, **W**.
5. Iodine solution.

*Before starting the investigation, read through steps **1-8** (including **(a)(i)**) and prepare a table in **(a)(ii)**.*

Proceed as follows:

1 Make a series of dilutions of **K** by **serial dilution**, diluting the concentration by a factor of 5.

2 You will need to repeat the dilution 4 times so that you have a range of 5 different concentrations. You are required to prepare for at least 5cm^3 of each concentration for step **3** onwards.

(a) **(i)** Complete the Table 1.1 to show how you will prepare the 5 different concentrations of **K**.

Table 1.1

Solution	Concentration of K / %	Volume of solution K taken from the previous dilution / cm^3	Volume of distilled water, W / cm^3	Total volume / cm^3
K1	5.0			8.0
K2	1.0	2.0	8.0	8.0
K3	0.2	2.0	8.0	8.0
K4	0.04	2.0	8.0	8.0
K5	0.008	2.0	8.0	10.0

[Mark Allocation]
1. All correct concentrations of K calculated correctly
2. Dilutions accurately calculated where all volumes when computed will result in correct concentration,

[2]

3 Set up a water bath with water between 38 to 42°C using the 500ml beaker. Monitor this temperature using the thermometer provided and maintain this temperature until you have completed step **8**.

4 Put 2 cm^3 of **S** into 6 separate test-tubes, then incubate these solutions in the water bath for at least 3 minutes.

5 Add 2 cm^3 of **K1** to one test-tube from step **4** and mix well. The reaction between **K** and **S** will begin as soon as **K1** is added.

6 Leave the solution to react in the water-bath for 5 minutes. Ensure that the solution mixture is mixed well throughout this 5 minutes.

7 After 5 minutes, add 0.5 cm^3 of the iodine solution into the test tube.

8 Repeat step **4** to **6** for all remaining concentrations of **K** you have prepared in step **1**. Record your results in **(a)(ii)**.

(ii) Record your results in a suitable table in the space below.

Solution	Concentration of K / %	Observed results of iodine solution test
K1	5.0	Iodine solution remained yellow
K2	1.0	Iodine solution turned from yellow slightly blue-black / Low amount of iodine solution turned from yellow to blue-black
K3	0.2	Some of iodine solution turned from yellow to blue-black
K4	0.04	Almost all of iodine solution turned from yellow to blue-black
K5	0.008	All of the iodine solution turned from yellow to blue-black

[Mark Allocation]

1. **Heading** – Correct headings, **including concentration of K** and **results of iodine solution test**, with unit % in the heading.
2. **Observations** – All **5** observations recorded, with **colour change** recorded instead of just stating final colour
3. **Trend** – **K1 should result in iodine solution remaining yellow** and **K5 should have all iodine solution turning blue-black**; increasing trend for K2 to K4.

[3]

9 Repeat step **4** to **6** for solution **P** and using your results from step **7**, estimate the concentration of **K** in the intestinal juice of the patient.

(iii) Estimation of concentration of **K** in sample **P**: …..……………0.04 to 0.2%………………..[1]

1. A range of answers accepted, as long as **<0.2%**; **unit** must be present

(no ECF from a(ii))

(iv) Hence, based on this concentration of **K** you have estimated, conclude with justifications whether the patient from which **P** is extracted is likely to have the genetic condition.

………

………[1]

1. Yes, as the **amount of starch remaining is high**, the **concentration of K is likely to be low** and hence the patient is likely to be carbohydrate intolerant.

(v) Given that the concentration of **K** for a patient with the genetic condition is around 0.05%, comment on the accuracy of your results in part **(a)(ii)** in helping you determine if the patient from which **P** is extracted has the genetic condition.

………

………

………

………[2]

1. The results can be considered accurate because all that is needed to determine if the patient has the genetic condition is to find out if there is a very low level of **K**, which this experiment is able to help discern the concentration of **K** appropriately.
2. However, the results could be inaccurate due **not being able to determine exactly how much K the patient has**;
3. since the **results for low levels** (0.2 and below) **of K were very similar** / **difficult to judge due to subjectivity of the intensity of blue-black colouration**
4. Because the **intervals of concentrations are few**, **not enough sensitivity to accurately assess** exact concentration

(vi) Suggest how the procedure from steps **1** to **2** could be modified to increase the accuracy of your results.

………

………[1]

1. Instead of serial dilution, perform **simple dilution** for a **range between 0.01 to 0.10**.

(vii) From the results of your investigation, deduce with rationale the identity of **K**, and suggest a control that you can perform to help you determine with more certainty the identity of **K**.

……

……

……

……………………………………………………………………………………………………….[2]

1. **K** is **amylase**; (accept enzyme), since the **concentration of starch decreases as the concentration of K increases.**
2. Use a **boiled solution containing K** and test for starch

(viii) Apart from ensuring that the solution mixtures were mixed well, identify two other aspects of the design of this investigation and explain how they are important in allowing valid comparisons to be made between the results in **(a)(ii)**.

……

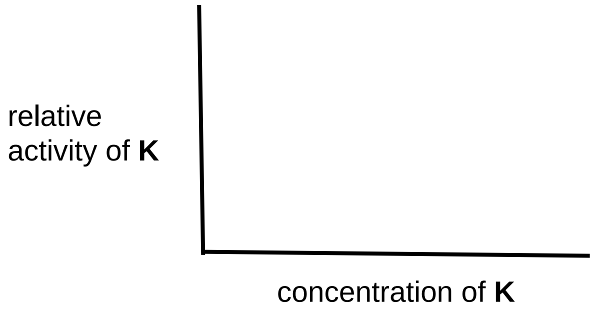
……

……

……………………………………………………………………………………………………….[2]

1. Temperature; since temperature **affects the activity of enzyme** to break down starch / accept: since **temperature will affect the rate** at which starch is broken down by K
2. Volume of starch solution; keeping the **concentration of the substrate constant**
3. Volume of iodine solution; allows for clearer differentiation between the different concentrations as compared to dropping a few drops / **standardised** volumes of **indicator**

(ix) The results of your investigation in **(a)(ii)** have allowed you to make qualitative comparisons between the various tubes. In order to complete a graph such as that shown below, it is necessary to obtain numerical values for the y (vertical) axis which give an indication of relative activity of **K**.



Outline how the procedure for this investigation could be further modified in order to obtain quantifiable numerical values for the relative activity of **K**.

………

………

………

……[2]

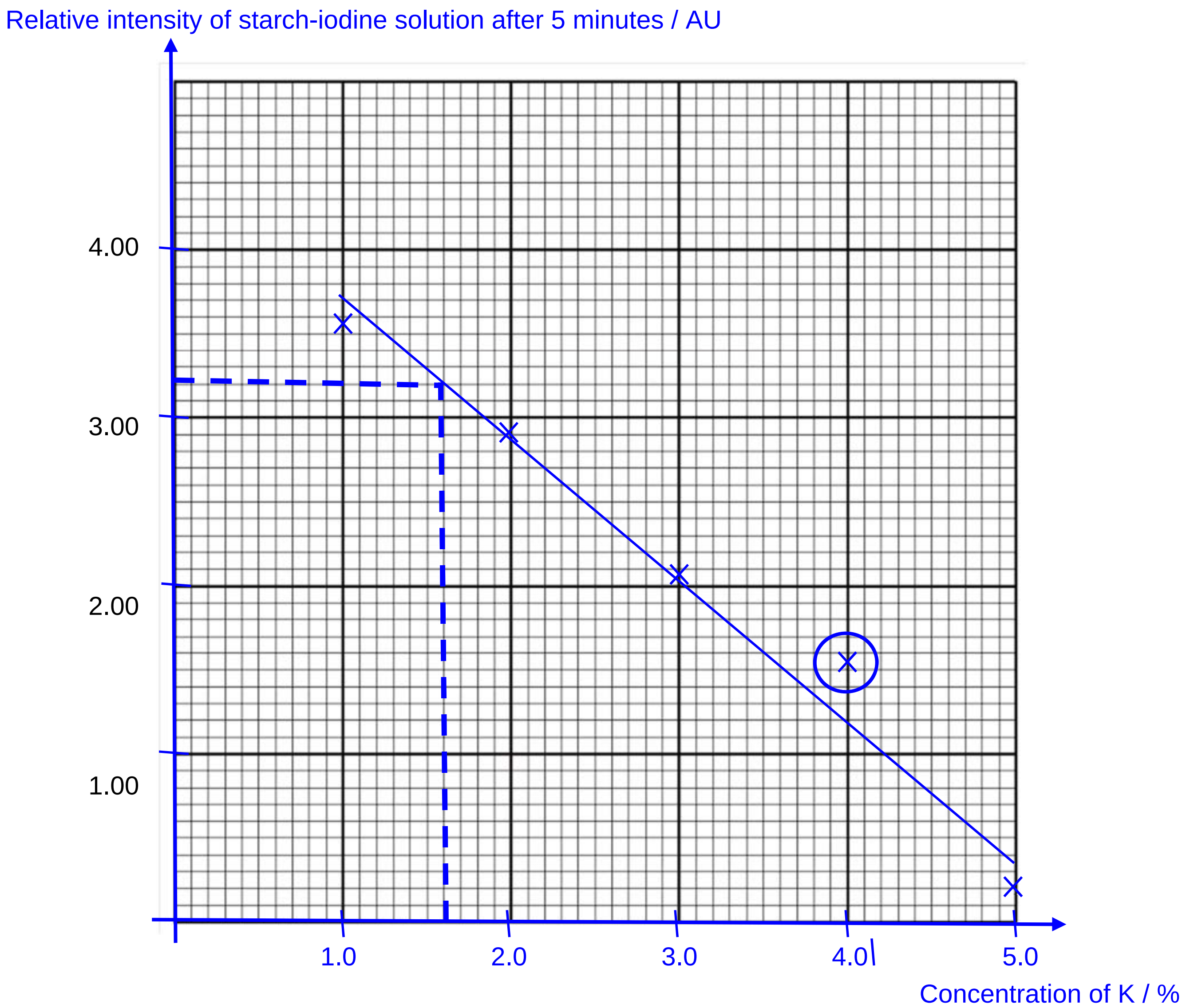
1. Set up **a range of known concentrations of starch solution** and perform the starch-iodine solution test;
2. **using a photospectrometer / colourimeter** to **measure** the **absorbance value** of the starch-iodine solution test instead of using human judgement to compare the relative concentration of starch remaining.

(b) A student carried out such a procedure to obtain quantifiable numerical values for the relative activity of **K**, using a similar procedure in **(a)** and measuring the intensity of the positive results of the starch-iodine solution test. The results for this investigation is shown in Table 1.2.

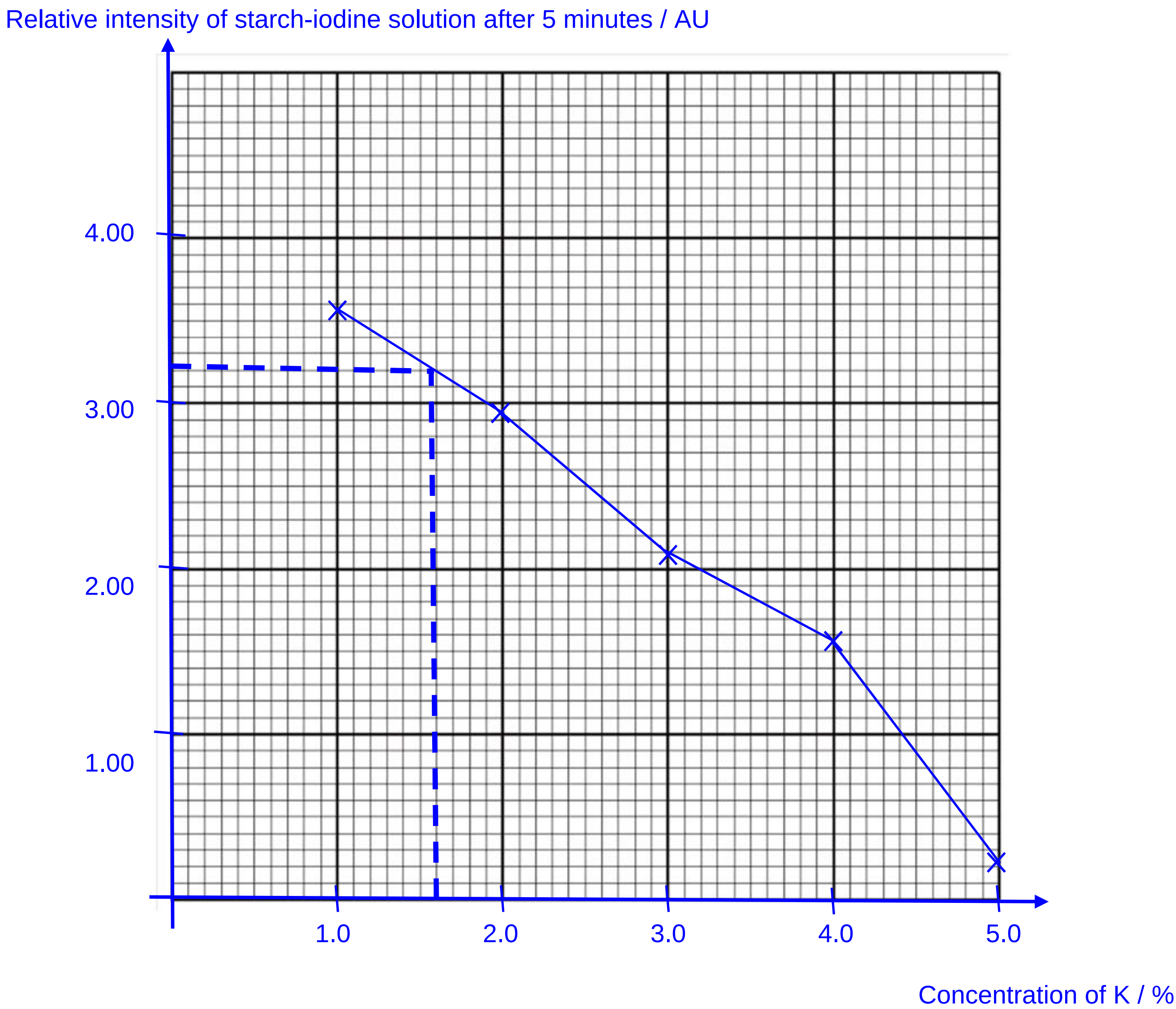
Table 1.2

Concentration of K / %	Relative intensity of starch-iodine solution after 5 minutes / AU (arbitrary unit)
5.0	0.20
4.0	1.55
3.0	2.05
2.0	2.90
1.0	3.55

(i) Use the grid below to display the student's results obtained in **(b)(i)** in an appropriate form.



[4]



[Mark Allocation]

[1] – **appropriate Scale** used for both axes, with **equidistant intervals / equally-spaced intervals;** graph to occupy at least **more than half of the given grid** (in both the x and y directions)

[1] – for **correctly labelled Axes with correct units** (Relative intensity of starch-iodine solution after 5 minutes / AU; Concentration of K / %)

[1] – **all points accurately Plotted** (within half a square) with **independent variable on the x-axis**

[1] – for **use of straight Line (i.e. line of best fit)** and **no extrapolation** beyond extreme plotted points, accept: use of line to join from dot to dot.
Reject: curved lines

(ii) Another patient B's intestinal juice sample produced a relative intensity of 3.20 AU with the starch-iodine solution test.

Show clearly how you would use your answer in **(b)(i)** to estimate the concentration of **K** in this patient.

…….

…….[1]

From the graph, 3.20 AU = 1.6% **K**; accept from graph
***Should not be able to estimate from a dot-to-dot graph

(iii) After setting up a negative control, the student measured a result of 4.00 AU (arbitrary unit) with the starch-iodine solution test. Assuming that the negative control resulted in 0.0 unit of relative activity of **K**, calculate the estimated relative activity of **K** in patient B. Show your working clearly.

4.01 AU (max substrate) = 0.0 relative activity

Hence, relative activity = 4.00 (max substrate) – x value (left over substrate)
[1m, accept for working shown below as substitute for conclusion**]**

3.20 AU = 4.00-3.20
= 0.80 relative activity **[1m]**

[2]

[Total: 23]

QUESTION 2

During this question you will require access to a microscope and slide **S1** and **S2**.

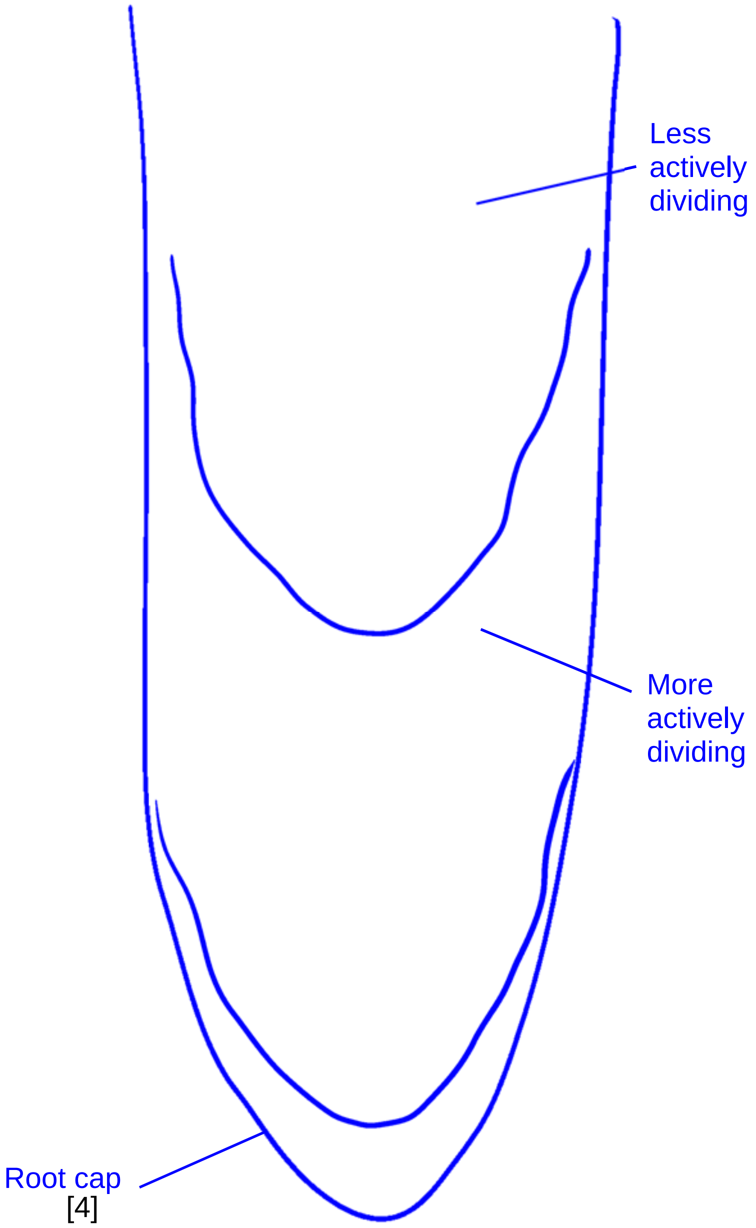
In this question, you will be investigating how cells appear during cellular division for different organisms.

(a) Slide **S1** shows a stained longitudinal section of a young root tip.

Examine **S1** carefully using low and high power objectives of your microscope. Note the occurrence and distribution of different kinds of cells in this section.

(i) Make a plan drawing of the entire section, to show the different **regions** of the root tip. These regions result from differences in the *shapes, sizes* and *structure* of the cells as well as in the frequency of mitosis. Do not draw individual cells.

Label in your diagram where the cap of the root tip is, and annotate your drawing to show the relative mitotic activity in each region that you map.



[4]

[Mark allocation]

Quality of drawing

Scale is appropriate to the space provided, **no cells drawn**.

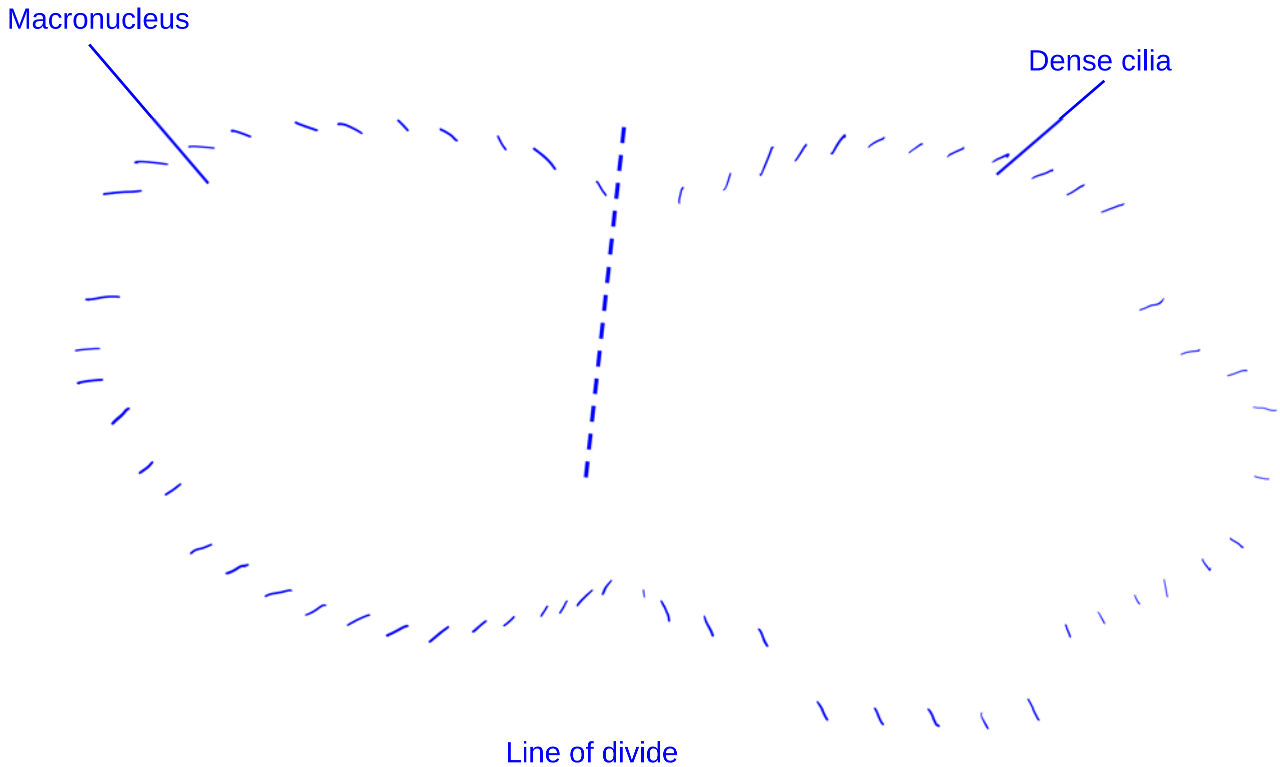
Accuracy of drawing

3 regions seen, with accurate relative proportion

Relative activity of the different region correctly

(b) Slide **S2** shows a unicellular eukaryote, *Paramecium*, that are usually found in freshwater bodies. They have two nuclei, a micronucleus that stores most of the genome, and a macronucleus which expresses the genes for daily functioning. Paramecium reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis. The cell then divides transversally, and each new cell obtains a copy of the micronucleus and the macronucleus.

(i) Look at slide **S2** under the high power objective. In the space below, make a drawing of a *Paramecium* that is clearly undergoing binary fission, where you can see two cells dividing from one. In your diagram, label one macronucleus as well as the line of divide.



[Mark Allocation]
Quality of drawing:

1. Clear continuous lines used for drawing cell
2. Appropriate scale, covering more than 50% of the space available

Accuracy of drawing:

3. Cells drawn in the middle of binary fission
4. Both macronuclei drawn clearly
5. Macronuclei and line of divide labelled clearly

[5]

(ii) Locate a cell undergoing **anaphase** in **S1**. Using a suitable table, record any observable differences you have noticed between the cell you have located in **S1** and the cell you have chosen to draw in **(b)(i)**.

Root tip cell undergoing anaphase	Paramecium undergoing binary fission
Chromosomes clearly seen	Macronuclei still exist as darkened region where chromosomes are not clearly defined
Overall shape of the cell remain rectangular / regular (due to cell wall)	Overall shape of the cell irregular
Presence of thick cell walls	Absence of cell wall; presence of hair like structure / cilia

1. **1 mark for each comparable feature that is different**

[2]

(iii) Using the stage micrometer to calibrate your microscope, show with working how much each eye piece graticule measures at objective lens of 40X.

1. Calculation of the eye piece graticule division:
   100 stage micrometer div = 1 cm
   1 stage micrometer div = 0.01 cm = 0.1 mm

2. @10X: 100 EPG div = 2.5 SM div = 0.25 mm = 250 µm
   Each eye piece graticule division = **2.5 µm**

[2]

(iv) Using your answer in **(iii)**, measure the size of the cell that you have drawn in **(i)** and show the magnification of your drawing.

1. Actual size of a dividing paramecium cell = Approx between 100 – 150 µm

2. Magnification = (value) **X**

[2]

(c) In a recent nitrate pollution of the school pond near the science laboratory, there was an increase growth in algae. A student decided to investigate whether the population size of *Paramecium* was affected by this pollution. In his investigation in growing *Paramecium* in two different types of water, he obtained a set of results shown in **Table 2.1**.

Table 2.1

Mean population size per 100 ml / cell count		Significance of difference	Total number of samples
Normal pond water	Polluted pond water		
530	891	$p < 0.05$	10

(i) State a statistical test that could be used to determine if the difference between the mean population sizes of *Paramecium* in the two different water bodies is significant.

……………………………………………………………………………………………………[1]

T-test

(ii) Based on the result seen in Table 2.1, explain what conclusions can be drawn for the results obtained by the student for the effect of nitrate pollution.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[2]

1. Since p < 0.05, there **is significant difference** between the population sizes in pond water and the polluted pond water

2. The pollution of nitrate has **significantly increased the population size / reproductive rate** of *Paramecium* in the pond

Accept: the pollution has a **significant effect** on the **population size / reproductive rate** of *Paramecium* in the pond

[Total: 18]

QUESTION 3

Bio-indicators refer to organisms being used to measure changes in the environment such as pollution level or ambient temperature, mainly because of how biotic factors can be taken into consideration which traditional instruments are not able to detect. Insects are a group of organisms that are frequently used as bio-indicators as they are abundant and are highly sensitive to even small changes in the environment due to their small sizes and natural morphologies. As insects have a very narrow physiological tolerance, climate change is going to affect these organisms even more so than other larger organisms.

As ectotherms (cold-blooded organisms), the rate of metabolism in insects are highly dependent on ambient temperature. The rate of respiration can be used to estimate the level of metabolism in insects. You are required to design an experiment to investigate how temperature affects the relative metabolic rates of crickets, an insect commonly found in Singapore.

Figure 3.1 shows an example of a set-up for a simple respirometer measuring respiration for seeds. Show in your plan how you would modify this set-up to measure the rate of respiration in crickets.

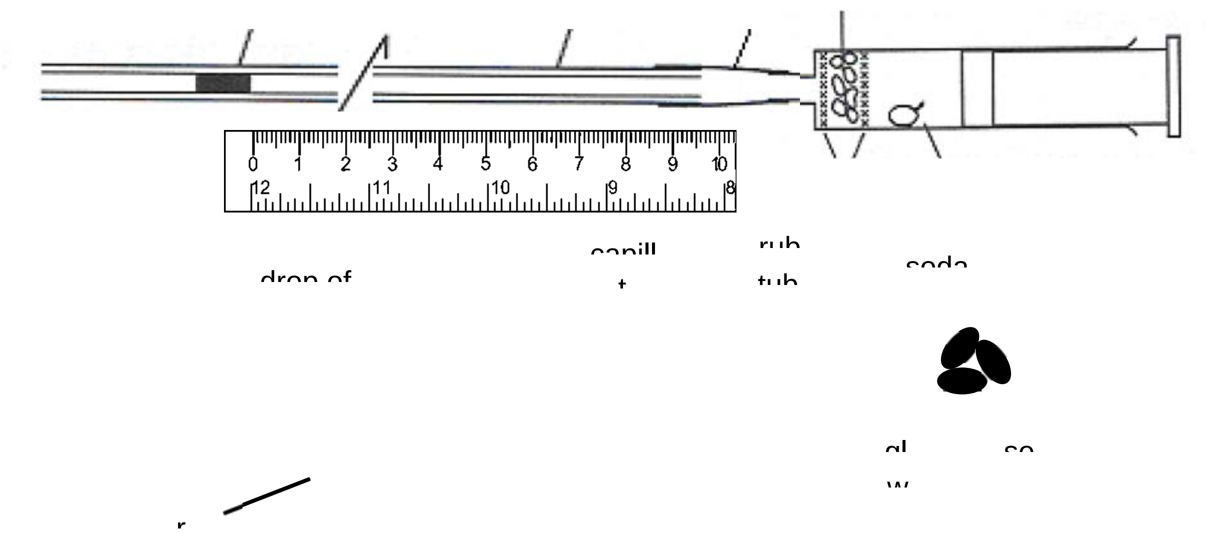


Fig. 3.1

In your plan, you must use:

- crickets
- soda lime
- delivery tube
- thermometer
- ruler

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- Bunsen burner
- timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagrams, if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

End of Paper

Question 3 ANS

AIM
To investigate how temperature affects the relative metabolic rates of crickets

HYPOTHESIS [1m]

1. Rate of aerobic respiration is used as an approximate to metabolic rate. **As temperature increases**, relative **rate of aerobic respiration of crickets first increases**, **then decreases as temperature increases further beyond optimal.**

Volume of oxygen is used to estimate the rate of respiration, and distance moved by the ink will increase as rate of respiration increases since more oxygen is taken in.

INTRODUCTION [1m]

2. Aerobic respiration utilises **oxygen as the final electron acceptor** and hence oxygen consumption increases as rate of respiration increases.

   As temperature increases, rate of respiration will increase due to **higher frequency of successful collision** (higher rate of diffusion) and **higher rate of enzyme-substrate formation**.

   At higher temperature after optimal temperature, rate of respiration decreases as **enzymes involved are denatured.**

VARIABLES [Max 6m]

3. Independent variable: **temperature**, using a range of **10°C to 50°C**

(5 values minimum, with a fixed and regular interval, units included)

4. changed by **mixing hot and cold water; measured by thermometer**
5. Dependent variable: **rate of respiration** indicated by volume of oxygen taken in

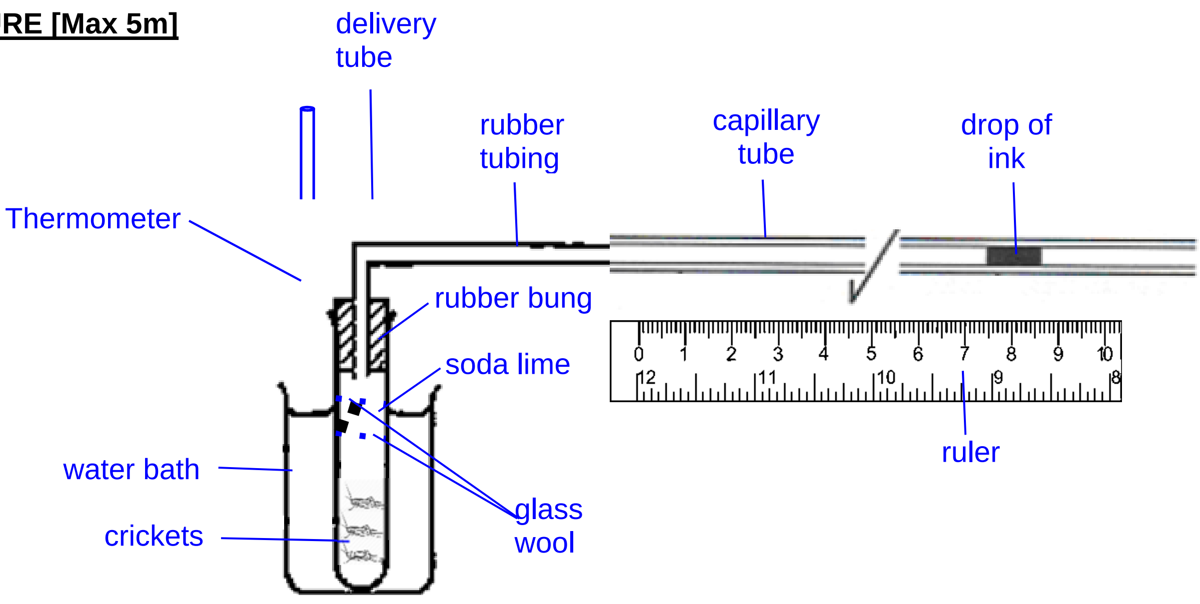
6. estimated by **distance moved by the drop of ink, in mm measured by a ruler** (accept: cm)

Controlled variables [max 2]

7. **Mass and number of crickets** used
   a. It affects the volume of oxygen absorbed / rate of respiration.
   b. Using a weighing balance, standardise the total mass of the organisms used for each reaction set-up at 20g; number of crickets should also be similar

8. **Duration of experiment**
   a. It affects the volume of gas measured
   b. A digital stopwatch/ timer is used to ensure the duration for respiration is kept constant at 5 minutes.
9. **Volume of drop of ink (AVP)**
   a. Affects how far the drop of ink moves due to weight
   b. Use a syringe to measure a consistent volume e.g. 0.1 cm^3

PROCEDURE [Max 5m]



10. Diagram (award 1m)

11. 1. Set up a water bath using a large / 500 ml beaker with hot and cold water with a thermometer.
2. Place **20g of crickets inside the boiling tube followed by glass wool, soda lime and then glass wool**. (accept: as shown in diagram, if diagram is drawn)
3. **Incubate** the boiling tube containing the crickets in the water bath **for at least five minutes to equilibrate.**

12. 4. Attach the delivery tube to the boiling tube using a rubber bung and ensure that it is air-tight, and attach the other end of the delivery tube to the capillary tube with the rubber tubing.
5. **Add in a drop of ink** using a small syringe, **only after the set-up is complete**.

13. 6. Start the digital stopwatch/ timer and time for one minute. (**after drop of ink is inserted**)
7. **Record the distance moved by the drop of ink using a ruler after one minute**.
(Take note: time should not be longer than 5 minutes, capillary tube is not long enough)

[Award mark for <u>logical</u> steps in <u>sequence</u> that ensure the experiment will work, up to 3m max]

14. 8. Repeat steps 1 to 7 twice using the **same** crickets to obtain a total of **three readings** for each temperature and determine the average distance moved by the drop of ink.

9. Repeat steps 1 to 8 for the other temperatures (no mark for this step)

15. 10. **Repeat the entire experiment** two more times using **fresh crickets** to **ensure reliability** of results obtained
(Award 1m max for 14 or 15)

16. 11. To show that the **oxygen absorbed is only due to the effect of respiration in crickets**, set up a negative control by **repeating steps 1 to 7 and 9**, but with only glass beads of the same mass.
To show that **temperature also had a negligible effect on the expansion of oxygen** and volume of oxygen absorbed is due to respiration.

[Turn over

SAFETY PRECAUTIONS [1m]

17. **Soda lime** is an irritant (hazard), **wear safety goggles** and **gloves** to avoid contact with eyes and skin (precaution)

18. **Handling hot water** (hazard), practise caution and **wear safety goggles** and **gloves** (precaution)

19. **Fragile glassware, such as thermometer and capillary tube** (hazard), practise caution and **handle these apparatuses with care** (precaution)

RECORDING OF RESULTS [Max 2m]

20. Drawing of a suitable table; **headings for independent and dependent variables** clear.

Table showing the average distance moved by drop of ink at different temperatures

Temperature / °C	**Distance moved by drop of ink / mm**			
	Reading 1	**Reading 2**	**Reading 3**	**Average**
10				
20				
30				
40				
50				

21. Showing of a suitable graph, **axes for indepdent and dependent variable** clear

Graph of average distance moved by drop of ink at different temperatures

Average distance moved by drop of ink / mm
Temperature /°C

INTERPRETATION [1m]

22. Distance moved by the ink is caused by the volume of oxygen taken in due to respiration. It increases initially due to increased rate of respiration as temperature increases, until optimal temperature is reached. Distance moved by the ink then decreases as temperature increases beyond optimal temperature since rate of respiration decreases due to denaturation of enzymes.

[Turn over